

Set 15: Product Sales Analysis

DSA – Data Handling and Visualization | Lab Experiment

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Aim

To analyse monthly product sales for the first quarter using a grouped bar chart, a stacked area chart and a labelled table, and to combine them in an interactive Tableau dashboard.

Software / Tools Used

R 4.3.3 with ggplot2 (charts), gridExtra (tables) and patchwork (dashboard layout); Tableau Desktop / Tableau Public (interactive dashboard).

Dataset

Product ID	Product Name	January Sales	February Sales	March Sales
1	Product A	2000	2200	2400
2	Product B	1500	1800	1600
3	Product C	1200	1400	1100

Program – Dataset Creation (common to all R tasks)

```
library(ggplot2)
library(gridExtra)
sales <- data.frame(
  Product_ID = c(1, 2, 3),
  Product_Name = c("Product A", "Product B", "Product C"),
  January_Sales = c(2000, 1500, 1200),
  February_Sales = c(2200, 1800, 1400),
  March_Sales = c(2400, 1600, 1100)
)
write.csv(sales, "Monthly_Product_Sales.csv", row.names = FALSE) # for Tableau

# Long format: one row per product per month
sales_long <- data.frame(
  Product = rep(sales$Product_Name, times = 3),
  Month = factor(rep(c("January", "February", "March"), each = 3),
    levels = c("January", "February", "March")),
  Sales = c(sales$January_Sales, sales$February_Sales, sales$March_Sales)
)
pal <- c("Product A" = "#0072B2", "Product B" = "#E69F00", "Product C" = "#009E73")
```

Q1. Grouped Bar Chart – Sales of Each Product in Quarter 1

Create a grouped bar chart to visualize the sales of each product for the first quarter. Label the chart elements.

R Program:

```
# Q1. Grouped bar chart - sales of each product in Q1
print(xtabs(Sales ~ Product + Month, data = sales_long))
q1_total <- aggregate(Sales ~ Product, data = sales_long, FUN = sum)
names(q1_total)[2] <- "Q1_Total"
cat("\nQuarter-1 totals:\n"); print(q1_total, row.names = FALSE)

p_grp <- ggplot(sales_long, aes(x = Month, y = Sales, fill = Product)) +
  geom_col(position = position_dodge(width = 0.8), width = 0.75) +
  geom_text(aes(label = format(Sales, big.mark = ","),
    position = position_dodge(width = 0.8), vjust = -0.4, size = 3.3) +
  scale_fill_manual(values = pal) +
  scale_y_continuous(limits = c(0, 2700), labels = scales::label_comma()) +
  labs(title = "Monthly Sales by Product - Quarter 1",
    x = "Month", y = "Sales", fill = "Product") +
```

```
theme_minimal(base_size = 12) + theme(plot.title = element_text(face = "bold"))
print(p_grp)
ggsave("s15_q1_grouped.png", p_grp, width = 7.5, height = 4.5, dpi = 150, bg = "white")
```

Console Output:

```
      Month
Product January February March
Product A    2000     2200   2400
Product B    1500     1800   1600
Product C    1200     1400   1100

Quarter-1 totals:
Product Q1_Total
Product A      6600
Product B      4900
Product C      3700
```

Output:

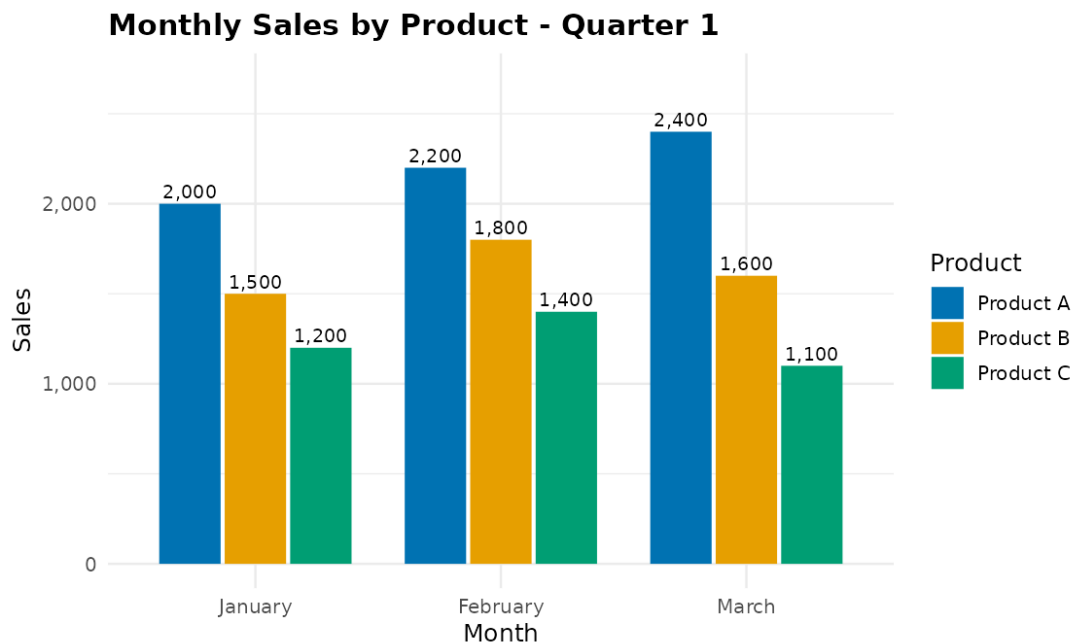


Figure 1: Grouped bar chart of monthly sales by product

Inference:

Product A is the top seller every month and is the only product that grows steadily (2,000 → 2,200 → 2,400), for a Q1 total of 6,600. Product B (total 4,900) and Product C (total 3,700) both peak in February and then fall in March.

Q2. Stacked Area Chart – Overall Sales Trend

Generate a stacked area chart to represent the overall sales trend for all products over the three months.

R Program:

```
# Q2. Stacked area chart - overall sales trend over three months
monthly_total <- aggregate(Sales ~ Month, data = sales_long, FUN = sum)
names(monthly_total)[2] <- "Total_Sales"
monthly_total$Change_pct <- c(NA, round(100 * diff(monthly_total$Total_Sales) /
                                                head(monthly_total$Total_Sales, -1), 1))
print(monthly_total, row.names = FALSE)

sales_long$Month_num <- as.numeric(sales_long$Month)
p_area <- ggplot(sales_long, aes(x = Month_num, y = Sales, fill = Product)) +
  geom_area(alpha = 0.85, colour = "white") +
  geom_text(data = monthly_total, aes(x = as.numeric(Month), y = Total_Sales,
                                      label = paste("Total:", format(Total_Sales, big.mark = ","))),
            inherit.aes = FALSE, vjust = -0.6, size = 3.6, fontface = "bold") +
  scale_fill_manual(values = pal) +
  scale_x_continuous(breaks = 1:3, labels = levels(sales_long$Month),
                    expand = expansion(mult = 0.14)) +
  scale_y_continuous(limits = c(0, 6000), labels = scales::label_comma()) +
  labs(title = "Overall Sales Trend - Stacked by Product",
```

```
x = "Month", y = "Sales", fill = "Product") +
  theme_minimal(base_size = 12) + theme(plot.title = element_text(face = "bold"))
print(p_area)
ggsave("s15_q2_area.png", p_area, width = 7.5, height = 4.5, dpi = 150, bg = "white")
```

Console Output:

Month	Total_Sales	Change_pct
January	4700	NA
February	5400	14.9
March	5100	-5.6

Output:

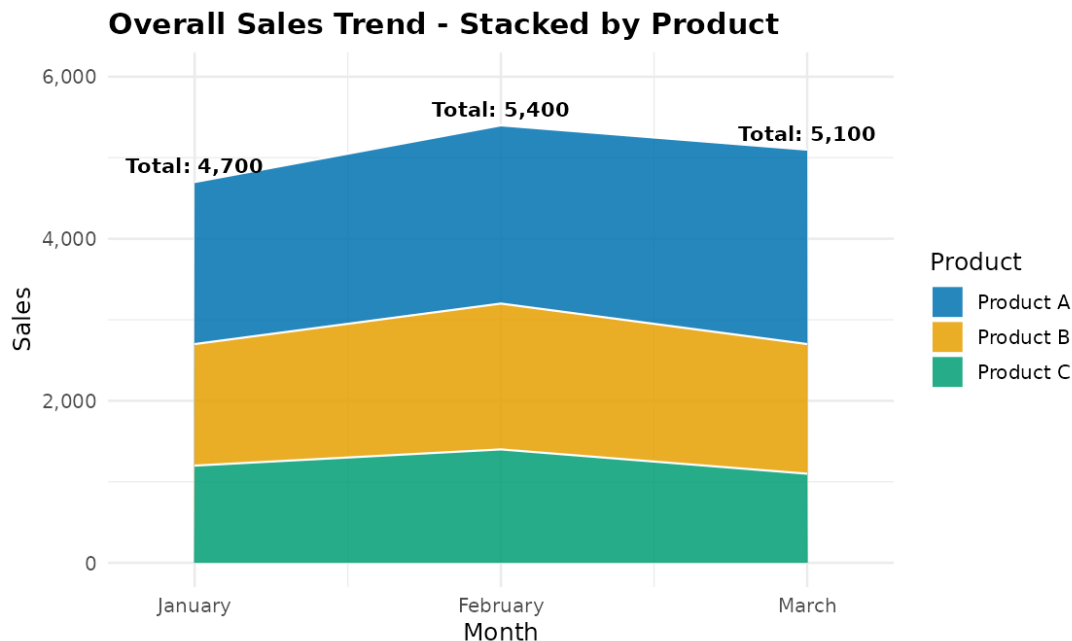


Figure 2: Stacked area chart of total sales by month

Inference:

Total sales rose by 14.9% from January (4,700) to February (5,400), then fell by 5.6% in March (5,100). Product A's growing layer offsets most of the March drop in Products B and C. Product A's share of total sales rises from 42.6% in January to 47.1% in March.

Q3. Table – Monthly Sales Data for Each Product

Build a table to show the monthly sales data for each product. Label the table elements.

R Program:

```
# Q3. Labelled table of monthly sales
tbl <- sales
tbl$Q1_Total <- rowSums(sales[, c("January_Sales", "February_Sales", "March_Sales")])
tbl <- rbind(tbl, data.frame(Product_ID = "", Product_Name = "Total",
  January_Sales = sum(sales$January_Sales),
  February_Sales = sum(sales$February_Sales),
  March_Sales = sum(sales$March_Sales),
  Q1_Total = sum(tbl$Q1_Total)))
names(tbl) <- c("Product ID", "Product Name", "January Sales", "February Sales",
  "March Sales", "Q1 Total")
print(tbl, row.names = FALSE)
tg <- tableGrob(tbl, rows = NULL, theme = ttheme_default(
  core = list(fg_params = list(fontsize = 11)),
  colhead = list(fg_params = list(fontsize = 11.5, fontface = "bold", col = "white"),
    bg_params = list(fill = "#1F3864"))))
title <- grid::textGrob("Monthly Sales Data by Product",
  gp = grid::gpar(fontsize = 14, fontface = "bold"))
png("s15_q3_table.png", width = 1200, height = 380, res = 150)
grid.arrange(title, tg, ncol = 1, heights = c(0.25, 1))
invisible(dev.off())
```

Console Output:

Product ID	Product Name	January Sales	February Sales	March Sales	Q1 Total
1	Product A	2000	2200	2400	6600
2	Product B	1500	1800	1600	4900
3	Product C	1200	1400	1100	3700
	Total	4700	5400	5100	15200

Output:

Monthly Sales Data by Product

Product ID	Product Name	January Sales	February Sales	March Sales	Q1 Total
1	Product A	2000	2200	2400	6600
2	Product B	1500	1800	1600	4900
3	Product C	1200	1400	1100	3700
	Total	4700	5400	5100	15200

Figure 3: Labelled monthly sales table with totals

Inference:

Quarter-1 sales total 15,200. Product A contributes 43.4%, Product B 32.2% and Product C 24.3%. February is the best month (5,400).

Q4. Tableau Dashboard – Interactive Exploration

Develop a Tableau dashboard combining the grouped bar chart, stacked area chart, and the table for interactive exploration of sales data.

R Program:

```
# Q4. Dashboard layout preview (Tableau dashboard built from Monthly_Product_Sales.csv)
library(patchwork)
dash <- (p_grp | p_area) / wrap_elements(full = tg) +
  plot_layout(heights = c(1.4, 0.6)) +
  plot_annotation(title = "Monthly Product Sales Dashboard",
    theme = theme(plot.title = element_text(face = "bold", size = 16)))
ggsave("s15_dashboard.png", dash, width = 13, height = 7, dpi = 130, bg = "white")
```

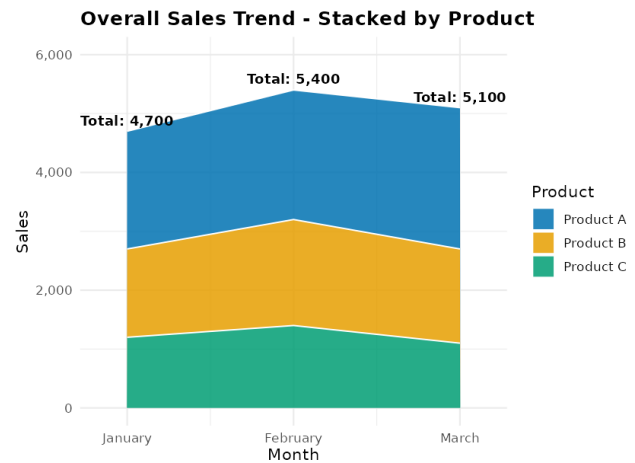
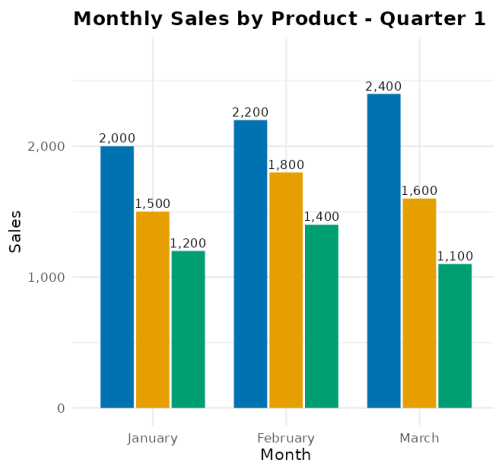
Procedure in Tableau:

- 1 Open Tableau → **Connect** → **Text file** → **Monthly_Product_Sales.csv**. In the Data Source tab, select January_Sales, February_Sales and March_Sales → right-click → **Pivot**. Rename the pivot fields to **Month** and **Sales**. Edit Month's aliases to January, February and March, and use a custom sort in month order.
- 2 **Sheet 1 – "Grouped Bar"**: drag Month to Columns, Product Name to Columns (to the right of Month) and SUM(Sales) to Rows. Drag Product Name to Color and show labels.
- 3 **Sheet 2 – "Stacked Area"**: drag Month to Columns and SUM(Sales) to Rows. Set Marks to **Area** and drag Product Name to Color, which stacks the areas by default. Add a reference line or label for the monthly total.
- 4 **Sheet 3 – "Sales Table"**: drag Product Name to Rows, Month to Columns and SUM(Sales) to Text. Turn on **Analysis** → **Totals** → **Show Row and Column Grand Totals**.
- 5 From the menu choose **Dashboard** → **New Dashboard**. Set **Size** → **Automatic** (or Fixed 1200 × 800).
- 6 Drag the sheets onto the canvas: Grouped Bar (top-left), Stacked Area (top-right), Sales Table (bottom). Add a **Text** object at the top for the dashboard title.
- 7 On each chart, click the sheet's drop-down and select **Use as Filter**. Clicking a bar or slice then filters the other views.
- 8 Go to **Dashboard** → **Actions** → **Add Action** → **Highlight** (Run on: Hover) so related marks highlight across all views.

- 9 Right-click a filter → **Apply to Worksheets** → **All Using This Data Source** (Monthly_Product_Sales) so one filter controls every view.
- 10 Save as a packaged workbook (**File** → **Save As** → **.twbx**) or publish to Tableau Public.
- 11 Add a **Product Name** quick filter (multiple values) so users can compare chosen products.

Dashboard Layout:

Monthly Product Sales Dashboard



Product ID	Product Name	January Sales	February Sales	March Sales	Q1 Total
1	Product A	2000	2200	2400	6600
2	Product B	1500	1800	1600	4900
3	Product C	1200	1400	1100	3700
	Total	4700	5400	5100	15200

Reference layout of the dashboard (rendered in R). The interactive version is built in Tableau using the procedure above.

Inference:

Selecting a product in the grouped bar chart isolates its layer in the area chart and its row in the table, which makes product-level trends easy to compare.

Result

The monthly product sales were visualised with a grouped bar chart, a stacked area chart and a labelled table in R, and combined into an interactive Tableau dashboard. Product A leads and grows every month. Total sales peaked in February (5,400), and the quarter total was 15,200.